UNIVERSITY OF TRANSPORT HO CHI MINH CITY

INSTITUTE OF INFORMATION TECHNOLOGY

FINAL TERM REPORT

Software Technology Learning Course

PROJECT TOPIC

PERSONALIZED CAREER ORIENTATION & LEARNING ROADMAP PLATFORM FOR SOFTWARE ENGINEERING STUDENTS

Group 2

Instructor	Nguyen Van Chien
Students	Nguyen Van Bao
	Phan Thi Duy Chung
	Vo Thi Kiem Dung
	Tran Thi Bich Diem
	Vo Trong Nguyen
	Vu Le Hoang Nhat
	Vo Bao Phuc

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I. Project Introduction

1. Overview

1.1 Project Information

- **Project name:** Personalized Career Orientation & Learning Roadmap Platform for Software Engineering Students
- **Project code:**
- **Group name:** Group 2
- **Software Type:** Web application

The rapid development of Information Technology has significantly transformed the way students learn, develop professional skills, and prepare for future careers. In particular, Software Engineering students are required to continuously update their technical knowledge, practical experience, and industry trends to remain competitive in the job market.

However, many students face difficulties in determining a suitable career path, identifying the required technical competencies, evaluating their current skill levels, and planning an effective learning roadmap. They often rely on scattered learning resources, inconsistent career advice, or personal experiences without systematic guidance. Consequently, students may spend considerable time learning technologies that are not aligned with their career goals or current industry demands.

The **Personalized Career Orientation & Learning Roadmap Platform for Software Engineering Students** is proposed to address these challenges. The platform provides intelligent career guidance by analyzing students' academic background, technical skills, career interests, and learning progress. Based on this information, the system generates personalized learning roadmaps, recommends appropriate learning resources, identifies missing competencies through skill gap analysis, and helps students build professional portfolios using their GitHub projects.

By integrating Artificial Intelligence, career planning, learning management, portfolio generation, and job market analysis into a unified platform, the proposed system aims to improve students' learning efficiency and enhance their career readiness.

1.2 Project Team

Table I.1: Project Team

Full Name	Role	Email	Mobile
Nguyen Van Bao	Project Manager		
Phan Thi Duy Chung	Business Analyst		
Vo Thi Kiem Dung	System Analyst		
Tran Thi Bich Diem	UI/UX Designer		
Vo Trong Nguyen	Backend Developer		
Vu Le Hoang Nhat	Frontend Developer		
Vo Bao Phuc	Tester		

2. Product Background

Software Engineering education focuses not only on theoretical knowledge but also on practical skills, project experience, and continuous learning. Modern software industries require graduates to possess diverse competencies, including programming, database management, software architecture, cloud computing, DevOps, artificial intelligence, and communication skills.

Despite the availability of numerous online learning platforms, students often encounter several challenges:

- Lack of personalized career guidance.
- Difficulty selecting an appropriate Software Engineering specialization.
- Limited understanding of industry-required technical skills.
- Inability to evaluate their own competencies objectively.
- Difficulty organizing learning plans according to career objectives.
- Lack of professional portfolios for internship and job applications.

Currently, students usually collect learning materials from various websites such as Microsoft Learn, Coursera, Udemy, roadmap.sh, GitHub, YouTube, and Stack Overflow. Since these resources are independent and not integrated, students must manually determine what to study, when to study, and how to evaluate their progress.

Therefore, a comprehensive platform that combines career orientation, personalized learning roadmap generation, AI-assisted mentoring, skill assessment, portfolio management, and labor market analysis can significantly improve students' learning outcomes and employment opportunities.

3. Existing Systems

Several existing platforms provide partial support for career planning or online learning; however, none of them fully integrate all required functionalities for Software Engineering students.

Roadmap.sh

Roadmap.sh provides structured technology roadmaps for different software engineering career paths, including Backend Development, Frontend Development, DevOps, AI Engineering, and Mobile Development. Although it offers detailed technology trees, it does not personalize learning paths based on students' individual competencies or academic performance.

Coursera

Coursera offers thousands of online courses from universities and technology companies. Students can obtain certificates after completing courses. However, course recommendations are primarily based on browsing history rather than comprehensive career analysis.

LinkedIn Learning

LinkedIn Learning provides professional courses and integrates with LinkedIn profiles. Nevertheless, it lacks personalized skill gap analysis and automatic roadmap generation tailored to Software Engineering students.

GitHub

GitHub serves as the largest source code hosting platform, allowing students to manage software projects and demonstrate programming experience. However, GitHub itself does not evaluate students' technical competencies or generate learning recommendations.

ChatGPT and Gemini

Artificial Intelligence chatbots such as ChatGPT and Gemini can answer programming and career-related questions. Nevertheless, they do not maintain long-term learning progress or automatically generate structured career roadmaps integrated with students' academic profiles.

Therefore, there is still a need for an integrated platform specifically designed for Software Engineering students.

4. Business Opportunity

The software industry continues to experience rapid growth worldwide. Companies increasingly demand graduates with practical technical skills rather than solely academic qualifications.

Universities are also shifting toward outcome-based education, emphasizing competency assessment, project-based learning, and career preparation. Consequently, educational institutions require systems capable of supporting students throughout their learning journeys.

The proposed platform provides several business opportunities:

- Supporting universities in career counseling.
- Assisting students in planning personalized learning paths.
- Helping employers evaluate candidates through dynamic portfolios.
- Assisting academic advisors in monitoring students' learning progress.
- Providing technology trend analysis based on labor market demands.
- Integrating Artificial Intelligence into educational support systems.

Furthermore, the platform can be expanded in the future to support multiple universities, online certification providers, internship management systems, and enterprise recruitment platforms.

5. Software Product Vision

The vision of the project is to become an intelligent career development platform that accompanies Software Engineering students throughout their academic journey and professional development.

The system aims to:

- Provide personalized career recommendations.
- Generate intelligent learning roadmaps.
- Continuously analyze students' skill gaps.
- Recommend high-quality learning resources.
- Build professional online portfolios automatically.
- Analyze current technology trends and employment demands.
- Improve students' employability and career readiness through data-driven recommendations.

By integrating Artificial Intelligence, GitHub repositories, academic information, and industry trends, the platform creates a comprehensive ecosystem for career development.

6. Project Scope & Limitations

6.1 Major Features

6.1.1 Major Features for Administrators

FE-01: Login/Logout.

FE-02: Manage user accounts (Students, Academic Counselors, Industry Mentors).

FE-03: Manage career paths (Backend Developer, Frontend Developer, Full Stack Developer, DevOps Engineer, AI Engineer, Data Engineer, Mobile Developer, QA Engineer, etc.).

FE-04: Manage skill categories and technical skills.

FE-05: Manage learning resources (courses, books, documentation, videos, practice websites).

FE-06: Manage AI prompt templates and system configurations.

FE-07: Manage job market data and technology trend information.

FE-08: Manage notifications and announcements.

FE-09: View system reports and analytics.

FE-10: Monitor user activities and system logs.

FE-11: Manage role-based access permissions.

6.1.2 Major Features for Student

FE-01: Register, login, logout, and authenticate using Google OAuth.

FE-02: Manage personal profile, including academic information, technical skills, career interests, and learning goals.

FE-03: Complete skill assessment to evaluate current competencies.

FE-04: Receive AI-powered career consultation through the AI Virtual Mentor.

FE-05: Select a target software engineering career path.

FE-06: Generate a personalized learning roadmap based on selected career goals.

FE-07: View and update learning progress for each roadmap milestone.

FE-08: Perform skill gap analysis to identify missing competencies.

FE-09: Receive recommended learning resources based on skill gaps.

FE-10: Connect and synchronize GitHub repositories.

FE-11: Automatically generate and maintain an online professional portfolio.

FE-12: View job market trends, technology demand, and career insights.

FE-13: Receive notifications about roadmap updates, learning recommendations, and career opportunities.

6.1.3 Major Features for Academic Counselor

FE-01: Login/Logout.

FE-02: View student profiles and academic information.

FE-03: Monitor students' learning roadmap progress.

FE-04: Review students' skill gap analysis reports.

FE-05: Provide academic advice and learning recommendations.

FE-06: Track students' learning performance and completion status.

FE-07: Send notifications or reminders to students.

FE-08: View reports summarizing students' learning progress.

6.1.4 Major Features for Industry Mentor

FE-01: Login/Logout.

FE-02: Review students' career roadmaps.

FE-03: Evaluate students' professional portfolios.

FE-04: Recommend industry-required technical skills.

FE-05: Provide career advice and industry feedback.

FE-06: Suggest learning priorities according to current market demand.

FE-07: Review students' project experiences and GitHub repositories.

FE-08: View students' competency development reports.

6.2 Limitations & Exclusions

LI-01: The first version of the system is designed exclusively for Software Engineering students and does not currently support students from other academic disciplines.

LI-02: AI career recommendations depend on external AI services (such as OpenAI or Gemini). Recommendation quality may vary depending on the AI model and available information.

LI-03: GitHub integration supports only repositories that are publicly accessible or explicitly authorized by users.

LI-04: Job market analysis depends on external recruitment platforms and publicly available datasets. Real-time accuracy cannot be fully guaranteed.

LI-05: The system currently supports predefined Software Engineering career paths only. Users cannot create custom career paths.

LI-06: The platform is implemented as a web application. Native Android and iOS applications are not included in this version.

LI-07: The platform requires a stable Internet connection because several core services depend on external APIs (AI services, Google Authentication, GitHub API, and job market data).

LI-08: The current version does not provide online coding practice, automatic code assessment, or integrated programming environments.

II. Project Management Plan

1. Overview

1.1 Project Scope and Estimation

The **Personalized Career Orientation & Learning Roadmap Platform for Software Engineering Students** is a web-based system designed to assist Software Engineering students in identifying suitable career paths and generating personalized learning roadmaps based on their academic performance, technical skills, interests, and career goals.

The project consists of the following major work packages.

Table II.1: Work Breakdown Structure (WBS) and Project Effort Estimation

WBS	Task	Complexity	Estimated Effort (Man-days)
1	Project Initiation	Medium	10
1.1	Requirement Gathering	Medium	5
1.2	Feasibility Analysis	Medium	5
2	Planning	Medium	8
2.1	Project Planning	Medium	4
2.2	Risk Planning	Medium	4
3	Analysis and Design	Complex	22
3.1	Requirement Analysis	Medium	8
3.2	Database Design	Medium	6
3.3	System Architecture	Complex	8
4	Implementation	Complex	80
4.1	Authentication Module	Simple	6
4.2	Student Profile Module	Medium	8
4.3	Career Assessment Module	Complex	12
4.4	Career Recommendation Engine	Complex	16
4.5	Learning Roadmap Generator	Complex	16
4.6	Learning Resource Management	Medium	10
4.7	Dashboard and Progress Tracking	Medium	12
5	Testing	Complex	24
5.1	Unit Testing	Medium	8
5.2	Integration Testing	Complex	8
5.3	System Testing	Complex	8
6	Deployment	Medium	8
6.1	Deployment	Medium	4
6.2	User Acceptance Testing	Medium	4

Total Estimated Effort: 152 Man-days

The objectives of the project are as follows.

- Develop a web-based personalized career orientation platform.
- Recommend suitable Software Engineering career paths.
- Generate personalized learning roadmaps.
- Track students' learning progress.
- Recommend appropriate learning resources.
- Improve students' career planning capability.
- Complete the project within one academic semester.

1.2 Testing Objectives

Table II.2: Testing Objectives

Testing Stage	**Coverage**	**Target Defects**
Code Review	Coding Standard	<10
Unit Testing	Business Logic	<8
Integration Testing	Module Communication	<5
System Testing	Complete Workflow	<5
Acceptance Testing	User Requirements	<3

1.3 Project Risks

Table II.3: Project Risks

Risk	**Impact**	**Probability**	**Mitigation**
Changing Requirements	High	Medium	Weekly meetings with supervisor
Implementation Delay	High	Medium	Sprint planning and monitoring
Incorrect Recommendation Results	Medium	Medium	Validation and testing
Database Performance Issues	Medium	Low	Query optimization
Communication Problems	Medium	Medium	Weekly meetings and GitHub tracking
Member Absence	High	Low	Knowledge sharing and backup assignments

2. Management Approach

2.1 Development Methodology

The project adopts the Agile Scrum methodology.
The reasons include:

- Continuous customer feedback
- Flexible requirement changes
- Incremental software delivery
- Continuous testing
- Better project monitoring

Each sprint lasts two weeks.

2.2 Quality Management

Quality assurance activities include:

- Requirement Review
- Code Review
- Unit Testing
- Integration Testing
- System Testing
- User Acceptance Testing

2.3 Training Plan

Table II.4: Training Plan

Technology	**Participants**	**Duration**
ASP.NET Core	All Members	1 Week
ReactJS	Front-end Developers	1 Week
SQL Server	All Members	3 Days
Git and GitHub	All Members	2 Days
Figma	UI Designer	2 Days

3. Project Deliverables

Table II.5: Project Deliverables

Sprint	Deliverables
1	Requirement Analysis
2	Database Design and UI Prototype
3	Authentication Module
4	Student Profile Module
5	Career Assessment Module
6	Career Recommendation Module
7	Learning Roadmap Module
8	Learning Resource Module
9	Dashboard and Progress Tracking
10	Integration Testing
11	System Testing
12	Deployment and Final Report

4. Responsibility Assignments

Table II.6: Responsibility Assignment Matrix

Task	Member 1	Member 2	Member 3	Member 4
Requirement Analysis	D	R	S	S
Project Planning	D	S	R	S
Database Design	R	D	S	S
Backend Development	S	D	R	S
Frontend Development	S	R	D	S
Testing	R	S	D	S
Documentation	S	R	S	D

Where

- D = Do
- R = Review
- S = Support

5. Project Communications

Table II.7: Project Communications

Communication	Participants	Frequency	Tool
Daily Meeting	Development Team	Daily	Google Meet
Sprint Planning	All Members	Every 2 Weeks	Google Meet
Sprint Review	Supervisor	Every Sprint	Offline/Meet
Issue Discussion	Related Members	As Needed	Discord
Document Sharing	All Members	Continuous	Google Drive
Source Code Review	Developers	Continuous	GitHub

6. Configuration Management

6.1 Document Management

The project documentation is maintained using:

- Microsoft Word
- Microsoft Excel
- Google Drive

6.2 Source Code Management

The source code is managed using GitHub and Git.

The following branching strategy is adopted:

- main
- develop
- feature/*
- release/*
- hotfix/*

6.3 Version Control Policy

- Every feature is developed in an independent branch.
- Pull Requests are mandatory before merging.

- Code review is required before approval.
- Semantic Versioning is applied.

Example versions:

- Version 1.0.0
- Version 1.1.0
- Version 2.0.0

III. Software Requirement Specification

1. Product Overview

The **Personalized Career Orientation & Learning Roadmap Platform for Software Engineering Students** is an intelligent web-based software platform designed to support Software Engineering students in planning their academic development and preparing for future careers. The system replaces the traditional approach of manually searching for career information, collecting learning resources from multiple websites, and creating self-designed learning plans without structured guidance. Instead, it provides a centralized platform that integrates AI-powered career consultation, personalized learning roadmap generation, skill gap analysis, portfolio management, and technology trend analysis into a single environment. The platform enables students to identify suitable career paths, monitor learning progress, build professional portfolios, and receive personalized recommendations based on their academic background, technical skills, and career aspirations.

The context diagram below illustrates the external entities and system interfaces for **Release 1.0** of the proposed platform. In this release, the system interacts with students, academic counselors, industry mentors, administrators, Artificial Intelligence services, GitHub, Google Authentication, and external job market data providers to deliver personalized career orientation and learning support. Future releases are expected to incorporate additional features such as internship recommendation, online certification integration, collaboration among students, mobile applications, and deeper integration with university information systems to provide a more comprehensive learning and career development ecosystem.

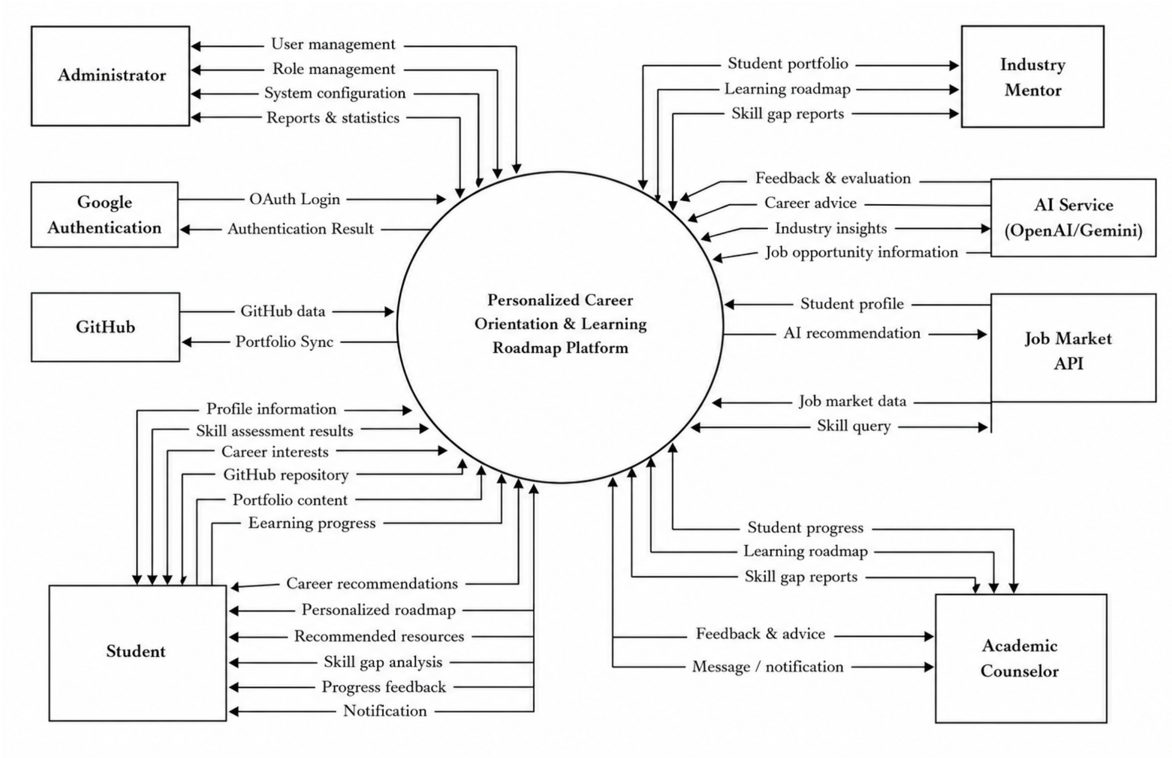

Figure III.1: Context Diagram

2. User Requirements

This section describes the users who interact with the system and the major use cases supported by the Personalized Career Orientation and Learning Roadmap Platform. These requirements define what each actor expects from the system and serve as the foundation for the functional requirements presented in the next section.

2.1 Actors

The system consists of internal users and external systems. Internal users directly interact with the platform, while external systems provide authentication, AI recommendation, repository analysis, and job information services. Table III.1 summarizes all actors involved in the system.

Table III.1: Actors of the System

No.	Actor	Description
1	Student	The primary user of the platform. Students complete career assessments, receive AI-based career recommendations, follow personalized learning roadmaps, enroll in courses, track learning progress, and apply for jobs or internships.
2	Career Counselor	Provides professional career guidance, reviews student profiles, monitors career development, recommends suitable learning paths, and communicates with students.
3	Mentor	Supports students during their learning journey by monitoring progress, answering questions, evaluating achievements, and providing technical guidance.
4	Administrator	Manages user accounts, learning resources, career information, system settings, reports, and ensures the secure operation of the platform.
5	AI Recommendation Engine	An external intelligent recommendation service that analyzes student profiles, skills, interests, and assessment results to generate personalized career recommendations and learning roadmaps.
6	Google OAuth	An external authentication service allowing users to log in securely using their Google accounts.
7	GitHub API	An external service used to retrieve students' repositories and programming activities for portfolio evaluation and skill analysis.
8	Job Portal API	An external recruitment platform providing internship and job opportunities that match students' career orientations.

2.2 Use Cases

This section describes the functional interactions between users and the Personalized Career Orientation and Learning Roadmap Platform. Each actor communicates with the system through a set of use cases that support career orientation, learning roadmap management, mentoring, administration, and external services.

The complete Use Case Diagram is presented in Figure III.2.

2.2.1 Use Case Diagram(s)

Figure III.2 illustrates the overall use case diagram of the Personalized Career Orientation and Learning Roadmap Platform. The diagram presents all primary actors and their interactions with the system.

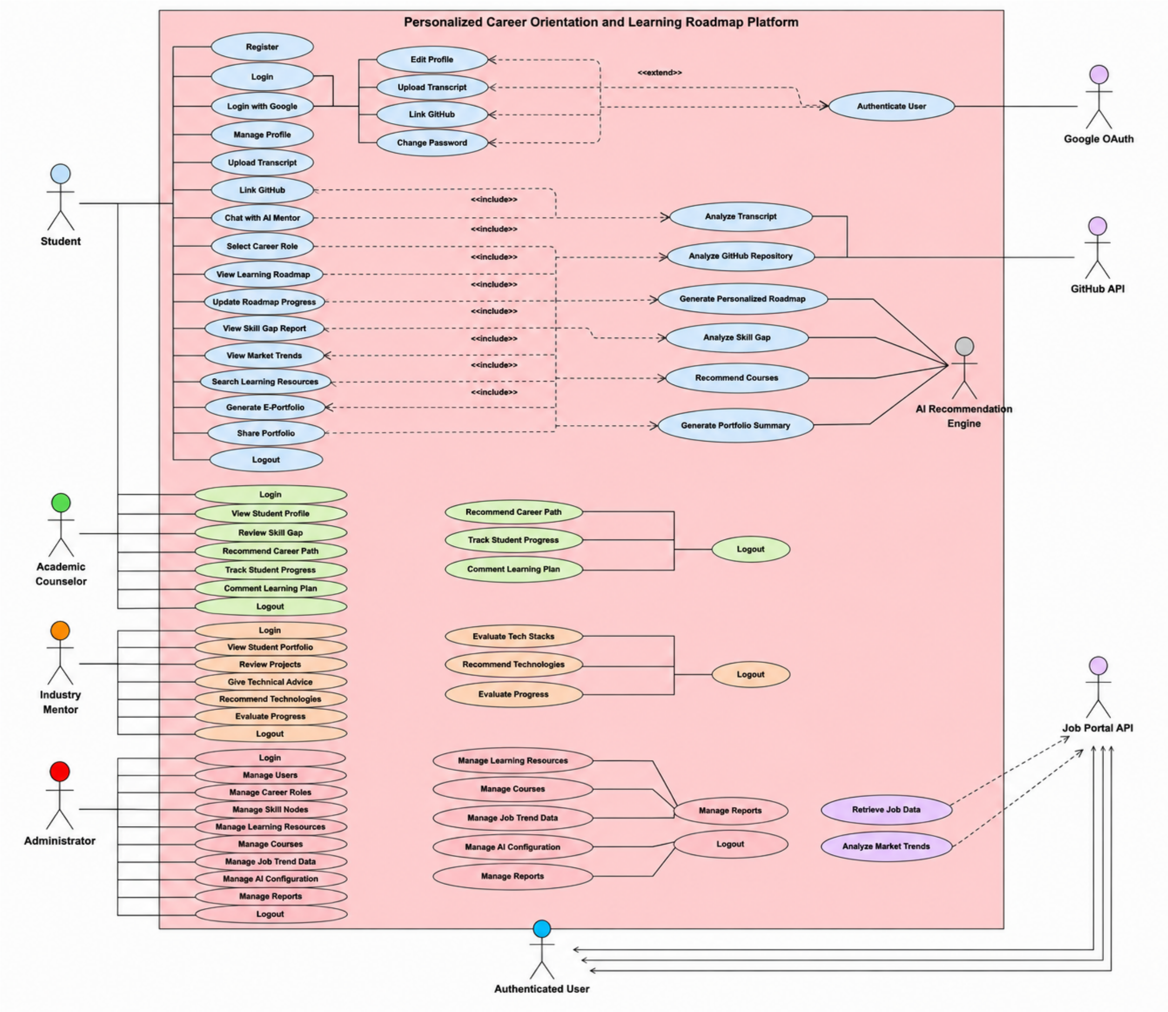

Figure III.2: Overall Use Case Diagram

2.2.2 Use Case Descriptions

Table III.2: Use Case Descriptions Summary

ID	Use Case	Brief Description	Actor
UC01	Register	Create a new student account.	Student
UC02	Login	Login using email and password.	Student
UC03	Login with Google	Authenticate via Google OAuth.	Student
UC04	Forgot Password	Recover forgotten password.	Student
UC05	Logout	Logout from the system.	Student

UC06	Manage Profile	Update personal information.	Student
UC07	Upload Avatar	Upload profile image.	Student
UC08	Update Contact Information	Edit phone number and address.	Student
UC09	Change Password	Update account password.	Student
UC10	Upload Transcript	Upload academic transcript.	Student
UC11	Upload CV	Upload curriculum vitae.	Student
UC12	Link GitHub	Connect GitHub account.	Student
UC13	Sync GitHub Repository	Synchronize repository information.	Student
UC14	View GitHub Analysis	View programming skill analysis.	Student
UC15	View Skill Gap Report	View missing skills analysis.	Student
UC16	Select Career Role	Select desired career path.	Student
UC17	Update Career Goal	Modify career objective.	Student
UC18	View Recommended Roles	View suggested careers.	Student
UC19	Compare Career Paths	Compare different career options.	Student
UC20	View Required Skills	View required competencies.	Student
UC21	Generate Learning Roadmap	Generate personalized roadmap.	Student
UC22	View Learning Roadmap	Display roadmap.	Student
UC23	Update Roadmap Progress	Update completed milestones.	Student
UC24	View Learning Progress	Monitor learning status.	Student
UC25	Receive Learning Reminder	Receive notifications.	Student
UC26	Search Learning Resources	Search learning materials.	Student
UC27	View Recommended Courses	View recommended courses.	Student
UC28	Save Favorite Courses	Bookmark courses.	Student
UC29	Register External Course	Enroll in external course.	Student
UC30	Track Course Completion	Track completed courses.	Student
UC31	View Market Trends	View labor market trends.	Student
UC32	Search Job Opportunities	Search job postings.	Student
UC33	View Salary Information	Display salary statistics.	Student

UC34	View Technology Trends	Display technology trends.	Student
UC35	Receive Job Recommendation	Receive recommended jobs.	Student
UC36	Chat with AI Mentor	Ask AI for learning advice.	Student
UC37	Ask Career Questions	Ask career-related questions.	Student
UC38	Receive AI Feedback	Receive AI feedback.	Student
UC39	Generate E-Portfolio	Generate portfolio automatically.	Student
UC40	Share Portfolio	Share portfolio link.	Student
UC41	Login	Login to counselor dashboard.	Academic Counselor
UC42	View Student Profile	View student profile.	Academic Counselor
UC43	Review Skill Gap	Review skill gap report.	Academic Counselor
UC44	Recommend Career Path	Recommend career orientation.	Academic Counselor
UC45	Recommend Courses	Recommend learning resources.	Academic Counselor
UC46	Track Student Progress	Track learning progress.	Academic Counselor
UC47	Comment Learning Plan	Provide comments.	Academic Counselor
UC48	Schedule Consultation	Arrange consultation session.	Academic Counselor
UC49	View Consultation History	Review consultation history.	Academic Counselor
UC50	Logout	Logout.	Academic Counselor
UC51	Login	Login to mentor dashboard.	Industry Mentor
UC52	View Student Portfolio	Review portfolio.	Industry Mentor
UC53	Review Projects	Review projects.	Industry Mentor
UC54	Recommend Technologies	Suggest technologies.	Industry Mentor
UC55	Give Technical Advice	Provide technical guidance.	Industry Mentor
UC56	Evaluate Progress	Evaluate learning outcome.	Industry Mentor

UC57	Comment Projects	Comment projects.	Industry Mentor
UC58	Approve Portfolio	Approve portfolio quality.	Industry Mentor
UC59	Suggest Improvements	Suggest improvements.	Industry Mentor
UC60	Logout	Logout.	Industry Mentor
UC61	Login	Login administrator.	Administrator
UC62	Manage Users	Manage user accounts.	Administrator
UC63	Manage Career Roles	Manage career roles.	Administrator
UC64	Manage Skill Nodes	Manage skill graph.	Administrator
UC65	Manage Courses	Manage courses.	Administrator
UC66	Manage Learning Resources	Manage learning resources.	Administrator
UC67	Manage Job Trend Data	Manage job market data.	Administrator
UC68	Manage AI Configuration	Configure AI engine.	Administrator
UC69	Manage Reports	Generate reports.	Administrator
UC70	Logout	Logout.	Administrator
UC71	Analyze Transcript	Analyze transcript.	AI Recommendation Engine
UC72	Analyze GitHub Repository	Analyze GitHub repository.	AI Recommendation Engine
UC73	Generate Personalized Roadmap	Generate roadmap.	AI Recommendation Engine
UC74	Analyze Skill Gap	Detect missing skills.	AI Recommendation Engine
UC75	Recommend Courses	Recommend learning resources.	AI Recommendation Engine
UC76	Generate Portfolio Summary	Summarize portfolio.	AI Recommendation Engine
UC77	Authenticate User	Authenticate user account.	Google OAuth
UC78	Retrieve Repository	Retrieve repository information.	GitHub API
UC79	Retrieve README	Retrieve README file.	GitHub API
UC80	Retrieve Tech Stack	Retrieve technology stack.	GitHub API
UC81	Retrieve Job Data	Retrieve job information.	Job Portal API
UC82	Analyze Market Trends	Analyze labor market trends.	Job Portal API

IV. Software Design

This chapter will be updated later.

V. Software Testing

This chapter will be updated later.

VI. Conclusion

This chapter will be updated later.